The On-Device AI Boom

MS Paint? Now upgraded with a text-to-image generator, thanks to AI. And prepare yourself, because soon, every app on your phone or laptop will incorporate AI. It's the buzzword, the mana fueling unforeseen advancements in the tech industry. Forget the cloud - the computing power for AI will reside on your device. Processors will boast AI capabilities, some purpose-built, others repackaged as "dedicated AI chips." It's the race for the ultimate AI-enabled processor that will find place in your personal computing device.

Cores are multiplying, and you'll find yourself tallying processor cores and units in various ways - from core processing CPUs to Neural Processing Units (NPUs) and Graphics Processing Units (GPUs). All collaborating seamlessly to tackle your evolving needs in this age of generative AI. On-device AI will demand faster, expanded memory and storage to bypass bottlenecks.

Intel and AMD are already gearing up for their consumer processor lineups, emphasizing an AI-first approach. The competition is heating up, with fresh benchmarks and stringent performance criteria taking the center stage. Meanwhile, Qualcomm's latest Snapdragon series focuses intensely on on-device AI computing in the smartphone realm. MediaTek's Dimensity chips are also playing the AI card prominently. Brace yourself, as generative AI permeates phones, laptops, and beyond.

Gaming? That's where AI is set to revolutionize. Picture this: every level, map, NPC, weapon, and power customized to your preferences and gaming style, all crafted by AI. Open World games will truly break boundaries. But all this creativity will demand serious AI computing power for real-time creation, rendering, and execution. And then comes Augmented Reality, where generative AI will craft virtual maps using real-world locations, blurring the line between reality and fantasy. Moreover, AI in gaming won't just customize things; it's going to change how games work. Imagine if the game could sense how you feel and change itself based on that, or adapt to how you play. AI might even become your in-game buddy, learning from you as you play. This means every time you game, it's a totally unique experience made just for you. The mix of AI and gaming isn't just about making things look cooler or different; it's about making games feel more personal and exciting than ever before.

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If chips are the new oil, then AI is the refinery, converting raw computational power into the innovation fueling every industry. The biggest challenge in advancing AI is making it efficient enough to save battery power. While processors are getting smarter with AI, batteries aren't keeping up. They're not improving as fast as the devices' power needs. The aim is to balance powerful AI with a battery that lasts. Engineers are trying new tricks, like smarter ways to manage energy, to find that perfect middle ground between strong performance and longer battery life. Getting this right is crucial for the future. It's not just about having a powerful device; it's about making sure it stays powerful without draining your battery too quickly. The devices that can do both – perform well with AI and keep your battery running longer – will be the real winners in the tech world.

I'm eager to witness how processors embrace this AI-first approach, unlocking new capabilities. We'll test and evaluate these innovations to provide a comprehensive guide on navigating the AI-driven world of devices, much like this month's issue showcasing the best-performing products of 2023 through our Zero1 Awards. Stay tuned for the next issue, brimming with accolades for on-device AI performance!

Will AI performance influence your choice of laptop or phone in 2024? Drop me a line at **Soham@digit.in**. I'd love to hear your thoughts!"